Exercise 1. (Squeeze Theorem) Let $S \subset \mathbb{R}$ and let c be a cluster point of S. Suppose $f : S \to \mathbb{R}$, $g : S \to \mathbb{R}$, and $h : S \to \mathbb{R}$ are functions such that

$$\forall x \in S : f(x) \leq g(x) \leq h(x).$$

Suppose the limits of $f(x)$ and $h(x)$ as x goes to c both exist, and

$$\lim_{x \to c} f(x) = \lim_{x \to c} h(x) \in \mathbb{R}.$$

Prove that the limit of $g(x)$ as x goes to c exists and

$$\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = \lim_{x \to c} h(x).$$

Proof. Let $S \subset \mathbb{R}$. Suppose $f, g, h : S \to \mathbb{R}$ are any real functions of S such that

$$\forall x \in S : f(x) \leq g(x) \leq h(x).$$

Assume c is a cluster point of S such that

$$\exists \lambda \in \mathbb{R} : \lim_{x \to c} f(x) = \lambda = \lim_{x \to c} h(x).$$

Suppose $\{x_n\}$ is any real convergent sequence of $S \setminus \{c\}$ such that

$$\lim_{n \to \infty} x_n = c.$$

By the Sequential Characterization of Limits,

$$\lim_{x \to c} f(x) = \lambda = \lim_{x \to c} h(x) \implies \lim_{n \to \infty} f(x_n) = \lambda = \lim_{n \to \infty} h(x_n).$$

Given $\{x_n \in S \setminus \{c\} \mid n \in \mathbb{N}\}$ is a proper subset of S,

$$\forall n \in \mathbb{N} : f(x_n) \leq g(x_n) \leq h(x_n).$$

By the Squeeze Theorem for Sequences,

$$\lim_{n \to \infty} g(x_n) = \lambda.$$

Then $\{x_n\}$ is an arbitrary convergent sequence of $S \setminus \{c\}$ such that

$$\lim_{n \to \infty} x_n = c \implies \lim_{n \to \infty} g(x_n) = \lambda.$$

By the Sequential Characterization of Limits,

$$\lim_{n \to \infty} g(x_n) = \lambda \implies \lim_{x \to c} g(x) = \lambda.$$

Thus,

$$\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = \lim_{x \to c} h(x) = \lambda,$$

as needed. □

Exercise 2. Let

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ 2x & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that f is continuous at $x = 0$ and discontinuous at $x = 1$.

Proof. Set

$$f(x) = \begin{cases} 0 & (x \in \mathbb{Q}) \\ 2x & (x \in \mathbb{R} \setminus \mathbb{Q}) \end{cases}.$$

We want to prove that

$$\lim_{x \to 0} f(x) = f(0).$$

Let $\epsilon > 0$. Choose $\delta = 2^{-1}\epsilon > 0$. If $x \in \mathbb{Q}$ and $0 < |x - 0| < \delta$, then

$$|f(x) - f(0)| = |0 - 0| = 0 < \epsilon.$$

If $x \in \mathbb{R} \setminus \mathbb{Q}$ and $0 < |x - 0| < \delta$, then

$$|f(x) - f(0)| = |2x - 0| = 2|x| < 2\delta = \epsilon.$$

We want to prove that

$$\lim_{x \to 1} f(x) \neq f(1).$$

Let $\epsilon = 1$ and $\delta > 0$. By the Density of $\mathbb{R} \setminus \mathbb{Q}$, $\exists x_0 \in \mathbb{R} \setminus \mathbb{Q}$ such that $2^{-1} \leq x_0 < 1 + \delta$. Choose $x = x_0$. Then

$$0 < x < 1 + \delta \implies 0 < |x - 1| < \delta$$

and

$$|f(x) - f(1)| = |2x - 0| = 2|x| \geq 1 = \epsilon.$$

□

Exercise 3. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Suppose $f(c) > 0$. Show that there exists an $\alpha > 0$ such that for all $x \in (c - \alpha, c + \alpha)$ we have $f(x) > 0$.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be any continuous function. Suppose $c \in \mathbb{R}$ and $f(c) > 0$. Then c is a cluster point of $\mathbb{R}$ and $f(c)$ is a positive value of $\mathbb{R}$. By the Limit Characterization of Continuity, we have

$$\lim_{x \to c} f(x) = f(c) > 0.$$

Choose $\epsilon \in (0, f(c)) \neq \emptyset$. Then $\exists \delta > 0$ $\forall x \in \mathbb{R}$ such that

$$|x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

Choose $\alpha \in (0, \delta] \neq \emptyset$. Then $\exists \alpha > 0$ $\forall x \in (c - \alpha, c + \alpha)$ such that

$$f(x) > f(c) - \epsilon > 0.$$

□

Exercise 4. Suppose $f : [-1, 0] \to \mathbb{R}$ and $g : [0, 1] \to \mathbb{R}$ are continuous and $f(0) = g(0)$. Define $h : [-1, 1] \to \mathbb{R}$ by $h(x) := f(x)$ if $x \leq 0$ and $h(x) := g(x)$ if $x > 0$. Show that h is continuous.

Proof. Let $f : [-1, 0] \to \mathbb{R}$ and $g : [0, 1] \to \mathbb{R}$ be any continuous functions such that $f(0) = g(0)$. Define $h : [-1, 1] \to \mathbb{R}$ by

$$h(x) = \begin{cases} f(x) & (x \leq 0) \\ g(x) & (x > 0) \end{cases}.$$

If $c \in [-1, 0)$, then

$$\lim_{x \to c} h(x) = \lim_{x \to c} f(x) = f(c) = h(c).$$

If $c \in (0, 1]$, then

$$\lim_{x \to c} h(x) = \lim_{x \to c} g(x) = g(c) = h(c).$$

If $c = 0$, then

$$\lim_{x \to c^-} h(x) = \lim_{x \to c^-} f(x) = f(c) = g(c) = \lim_{x \to c^+} g(x) = \lim_{x \to c^+} h(x) = h(c).$$

□

Exercise 5. Let $f : \mathbb{R} \to \mathbb{R}$. Recall that if $U \subset \mathbb{R}$, the inverse image of U is the set

$$f^{-1}(U) := \{x \in \mathbb{R} : f(x) \in U\}.$$

Prove that f is continuous if and only if for every open set $U \subset \mathbb{R}$, $f^{-1}(U)$ is open.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$. We want to prove that

$$f \text{ is continuous } \implies \forall \text{ open } U \subset \mathbb{R} : f^{-1}(U) \text{ is open.}$$

Suppose f is continuous and $U \subset \mathbb{R}$ is open. If $x_0 \in f^{-1}(U)$, then $f(x_0) \in U$ and

$$\exists \epsilon_0 > 0 \ \forall y \in (f(x_0) - \epsilon_0, f(x_0) + \epsilon_0) : y \in U;$$

hence,

$$\exists \delta_0 > 0 \ \forall x \in (x_0 - \delta_0, x_0 + \delta_0) : |f(x) - f(x_0)| < \epsilon_0 \implies f(x) \in U$$

and $x_0 \in f^{-1}(U)^{\circ}$. Thus, $f^{-1}(U) \subset \mathbb{R}$ is open, as needed. We want to prove that

$$\forall \text{ open } U \subset \mathbb{R} : f^{-1}(U) \text{ is open } \implies f \text{ is continuous.}$$

Conversely, suppose $\forall$ open $U \subset \mathbb{R}$, $f^{-1}(U) \subset \mathbb{R}$ is open. If $x_0 \in \mathbb{R}$, then

$$\forall \epsilon > 0 : (f(x_0) - \epsilon, f(x_0) + \epsilon) \text{ is open } \implies f^{-1}[(f(x_0) - \epsilon, f(x_0) + \epsilon)] \text{ is open}$$

and

$$\exists \delta_0 > 0 : (x_0 - \delta_0, x_0 + \delta_0) \subset f^{-1}[(f(x_0) - \epsilon, f(x_0) + \epsilon)];$$

hence,

$$\forall \epsilon > 0 \ \exists \delta_0 > 0 \ \forall x \in \mathbb{R} \text{ such that } |x - x_0| < \delta_0 \implies |f(x) - f(x_0)| < \epsilon.$$

Thus, f is continuous, as needed. □